Comparison of AWS IoT Core and Azure IoT Hub

1. Limitations Comparison

Message Processing Capacity

Feature	AWS IoT Core	Azure IoT Hub
Maximum messages per connection	100 messages/second	Depends on tier (S1: 100/sec, S2: 120/sec, S3: 6,000/sec)
Maximum bandwidth per connection	512 KB/second	Depends on tier
Maximum message size	128 KB	256 KB (device-to-cloud), 64 KB (cloud-to-device)
Maximum concurrent connections	Unlimited (no practical limit)	Depends on tier
New connection rate	3,000/second per account (default)	Depends on tier (S1: 100/sec or 12/sec/unit, S2: 120/sec/unit, S3: 6,000/sec/unit)

AWS IoT Core is limited to 100 messages/second per connection, and this limit cannot be increased. It also has a bandwidth limit of 512 KB/second per connection. This can create a potential bottleneck in scenarios requiring high-volume message transmission.

Azure IoT Hub can offer higher message processing capacity depending on the tier. Especially in the S3 tier, it can process 6,000 messages/second per unit, and this capacity can be increased by using multiple units.

Protocol Support

Protocol	AWS IoT Core	Azure IoT Hub
MQTT	✓	✓
MQTT over WebSocket	✓	✓
HTTPS	✓ (publish only)	✓
AMQP	✗	✓
AMQP over WebSocket	✗	✓

Azure IoT Hub offers more protocol support compared to **AWS IoT Core**. Especially AMQP protocol support can provide an advantage in some enterprise scenarios.

2. Pricing Comparison

AWS IoT Core Pricing Structure

AWS IoT Core pricing is based on five main components:

1. **Connectivity**: Charged for the time your devices are connected to AWS IoT Core. Connection time is measured in minutes.
   - Example: $0.08/1,000,000 minutes of connection in the US East (N. Virginia) Region
   - Annual cost for 24/7 connectivity: Approximately $0.042/year per device
2. **Messaging**: Charged based on the number of messages transmitted between your devices and AWS IoT Core.
   - Messages are metered in 5 KB increments (an 8 KB message is metered as two messages)
   - Price: Varies by region, typically around $1 per million messages
3. **Device Shadow and Registry**: Charged for operations that store device state or metadata.
   - Operations are metered in 1 KB increments
4. **Rules Engine**: Charged for rules used to transform and route device data.
   - Rules and actions are metered in 5 KB increments
5. **Device Location**: Charged for location solutions.

Azure IoT Hub Pricing Structure

Azure IoT Hub pricing is based on the selected tier and number of units:

1. **Tiers**:
    - S1: 400,000 messages/day per unit, $25/month per unit
    - S2: 6,000,000 messages/day per unit, $250/month per unit
    - S3: 300,000,000 messages/day per unit, $2,500/month per unit
2. **Units**: Multiple units can be purchased for each tier, increasing the total capacity.
3. **Included Features**: Pricing includes device identities, messaging, device twins, and command-control functions.

Example Scenario Comparison (50,000 and 100,000 devices)

Scenario: Year 1: 50,000 devices, Year 2: +50,000 devices (total 100,000 devices). Each device sends an average of 200 messages per day.

AWS IoT Core Cost:

1. **Connectivity**:
    - Year 1: 50,000 devices * $0.042/year = $2,100
    - Year 2: 100,000 devices * $0.042/year = $4,200
2. **Messaging**:
    - Year 1: 50,000 devices * 200 messages/day * 365 days = 3.65 billion messages/year
        - 3.65 billion messages * $1/million messages = $3,650
    - Year 2: 100,000 devices * 200 messages/day * 365 days = 7.3 billion messages/year
        - 7.3 billion messages * $1/million messages = $7,300
3. **Total Estimated Cost**:
    - Year 1: $2,100 + $3,650 = $5,750
    - Year 2: $4,200 + $7,300 = $11,500
    - 2-Year Total: $17,250

Azure IoT Hub Cost:

1. **S2 Tier**:
    - Year 1: 10,000,000 messages/day ÷ 6,000,000 messages/day/unit = 1.67 units (rounded to 2 units)
        - 2 units * $250/month * 12 months = $6,000
    - Year 2: 20,000,000 messages/day ÷ 6,000,000 messages/day/unit = 3.34 units (rounded to 4 units)
        - 4 units * $250/month * 12 months = $12,000
    - 2-Year Total: $18,000

> In this scenario, the 2-year total costs of **AWS IoT Core** and **Azure IoT Hub** are quite similar. **AWS IoT Core**'s component-based pricing can provide an advantage in scenarios with low connection time and high message volume. **Azure IoT Hub**'s tier-based pricing offers more predictable costs and may make it easier to integrate additional features (such as Device Update, Defender for IoT).

3. Security Features Comparison

AWS IoT Device Defender vs Azure Defender for IoT

Feature	AWS IoT Device Defender	Azure Defender for IoT
Device auditing	✓	✓
Continuous monitoring	✓	✓
Anomalous behavior detection	✓ (ML-based)	✓ (ML-based)
Vulnerability management	✓	✓
OT/ICS device support	**Limited**	**Comprehensive**
Network traffic analysis	**Limited**	**Comprehensive**
Integration	With AWS services	With Azure services
Deployment model	Cloud only	Cloud and on-premises

AWS IoT Device Defender

AWS IoT Device Defender is a service designed to audit and monitor the security of devices connected to AWS IoT Core. It specifically provides:

- **Auditing**: Checks the security configuration of your IoT device fleet and compares it with best practices.
- **Detection**: Uses machine learning models to detect abnormal device behaviors.
- **Metrics**: Monitors 13 different metrics for both cloud-side (e.g., authorization failure counts, message sent counts) and device-side (e.g., outgoing packets, listening TCP port

counts).
- **Automatic Learning**: Automatically learns behavior patterns using ML models with device data from the last 14 days, without the need to create manual rules to define device behaviors.

Azure Defender for IoT

Azure Defender for IoT provides comprehensive security for both cloud-based IoT solutions and operational technology (OT) networks:

- **Comprehensive Coverage**: Supports not only IoT devices but also OT devices such as industrial control systems (ICS) and SCADA systems.
- **Network Monitoring**: Analyzes device traffic with passive network monitoring and detects anomalies.
- **Threat Intelligence**: Integrated with Microsoft's threat intelligence.
- **On-Premises Solution**: Can work in environments without cloud connectivity.
- **Zero Touch**: Can monitor existing devices without requiring agents.

Key Differences

1. **Scope**: **Azure Defender for IoT** has a broader scope than **AWS IoT Device Defender**, especially offering more comprehensive support for industrial systems and OT devices.
2. **Deployment Model**: **Azure Defender for IoT** offers both cloud and on-premises deployment options, while **AWS IoT Device Defender** is cloud-only.
3. **Integration**: Both solutions provide deep integration with their respective cloud ecosystems. **AWS IoT Device Defender** integrates with AWS IoT Core and other AWS services, while **Azure Defender for IoT** integrates with Azure IoT Hub and other Azure security services.
4. **Network Monitoring**: **Azure Defender for IoT** has stronger features in network traffic analysis and is deeper in protocol analysis.

4. Overall Assessment

Advantages of AWS IoT Core

1. **Flexibility**: Component-based pricing offers pay-as-you-go capability.
2. **AWS Ecosystem**: Easy integration with other AWS services.
3. **High Scalability**: Practically unlimited device connections.
4. **Simple Start**: Easier to start with a simpler structure.

Advantages of Azure IoT Hub

1. **Higher Message Processing Capacity**: 6,000 messages/second per unit in the S3 tier, higher than AWS's limit of 100 messages/second per connection.
2. **More Protocol Support**: Support for AMQP and AMQP over WebSocket.
3. **Larger Message Size**: 256 KB for device-to-cloud (128 KB in AWS).
4. **Comprehensive Security**: **Azure Defender for IoT** offers more comprehensive security features, especially in industrial IoT scenarios.

Recommendations for Your Scenario

For your specified scenario (50,000 devices in year 1, 100,000 devices in year 2, 200 messages per device per day):

1. **Message Processing Capacity**: While your daily message volume is high, the number of messages processed per second is a critical factor. If messages are evenly distributed throughout the day, both platforms can handle this load. However, if messages are concentrated in specific time periods, **Azure IoT Hub**'s higher message processing capacity may provide an advantage.
2. **Cost**: The 2-year total cost of both platforms is similar. **AWS IoT Core**'s component-based pricing may be more flexible, while **Azure IoT Hub**'s tier-based pricing is more predictable.
3. **Security**: If you have industrial IoT devices or are looking for comprehensive security features, **Azure Defender for IoT** may be a better option. **AWS IoT Device Defender** provides adequate security for standard IoT scenarios.
4. **Ecosystem**: The cloud platform you are already using may influence your choice. If you are using the AWS ecosystem, **AWS IoT Core** will provide better integration; if you are using the Azure ecosystem, **Azure IoT Hub** will provide better integration.

> In conclusion, both platforms are technically sufficient for your specified scenario. Your choice may depend on your existing cloud strategy, security requirements, and message processing patterns.